

RAN Architecture Options and Performance for 5G Network Evolution

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Abstract — This paper discusses various RAN architecture design options, from fully centralized cloud-RAN to fully distributed edge cloud, as candidates for 5G network evolution. System-level performance is evaluated for each architecture option, while we also outline our vision for 5G network performance based on our extensive simulation studies. We further discuss the role of network virtualization, its challenges, and the expected technical and economic benefits for implementing the different RAN architecture options for 5G networks.

Keywords — RAN re-architecture; CRAN; Anchor-Booster; Edge cloud; 5G; network virtualization; SDN; NFV; cooperative networking

I. INTRODUCTION

Mobile traffic has been growing explosively as new data-centric mobile devices such as smartphones and tablets have become popular. Compared to 2011, global mobile traffic grew 70% in 2012 [1]. On the other hand, the cost to build, upgrade and expand the mobile broadband network infrastructure is becoming higher. Therefore, the challenges presented to mobile network designers can be overcome with new approaches to RAN architecture evolution.

Various RAN architecture options have been proposed, ranging from centralized RAN to fully distributed RAN. Centralized RAN solutions with ideal backhaul are 3GPP standard compliant and, thus, enable faster time to market. Solutions like anchor-booster and edge cloud are designed with non-ideal backhaul in mind, and need further standardization work to ensure interoperability. Typically, different RAN architecture options have their own desired deployment scenarios, and the operator is required to flexibly tailor its deployment to the different options depending on the specific scenario. Network virtualization promises freedom from the specific deployment scenario and RAN architecture option by enabling implementation of various functionalities using the same high-performance platforms.

The paper is organized as follows. In Section II, we discuss various RAN architecture options with their advantages and limitations. In Section III, system-level performance results are presented for the evaluated options, including our long-term vision for 5G network performance. In Section IV, network function virtualization which is essential to enable the new RAN architecture is discussed. Session V summarizes our key observations and results and concludes the paper

II. RAN ARCHITECTURE OPTIONS

A. Centralized RAN

The Centralized, Co-operative and Cloud RAN (C-RAN) concept has been promoted by China Mobile Research Institute (CMRI) [2]. The C-RAN architecture enjoys significant benefits across the following key areas:

- **Energy efficiency:** The power consumption of air conditioning and other site support equipment can be largely reduced due to the consolidation of baseband unit (BBU) processing.
- **Capital expenditure (CAPEX) and operational expenditure (OPEX):** The pooling of BBUs allows centralized management and operation, leading to significant cost savings on site rental as well as operation and maintenance.
- **Capacity:** More advanced transmit and receive wireless communications technologies such as cooperative multi-point processing (CoMP) can be implemented using the C-RAN infrastructure, leading to remarkable network capacity improvements.
- **Traffic offloading:** BBU pooling can serve as a natural local breakout point to offload core network traffic to different network nodes as well as different radio access technologies if available.

Fig. 1 shows the typical deployment scenario originally considered for C-RAN. The BBU pool is placed at a central/regional location, thus enabling a large scale of resource and interference coordination. The deployment is also 3GPP compliant, which is expected to reduce the time to market of C-RAN solutions. With increasing interest in small cell heterogeneous network (HetNet) deployments, the local C-RAN deployment scenario illustrated in Fig. 2 is becoming more popular. In local C-RAN, the BBU pool is located at the macro-cell site. The remote radio heads (RRHs) are connected to the BBU pool through high-rate fronthaul links. The localized nature of this scenario can enable the deployment of C-RAN in densely populated indoor deployments such as offices, high rise apartment buildings and stadiums. It is also possible to use existing wiring in the building, such as Ethernet cable, as the fronthaul link when proper re-

architecture and compression technologies are applied, as we discuss later in this section.

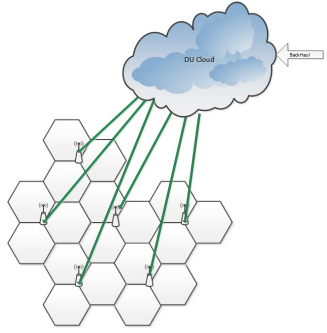


Fig. 1. BBU fully centralized (original C-RAN approach)

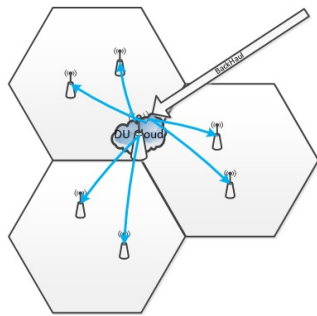


Fig. 2. BBU locally centralized (local C-RAN)

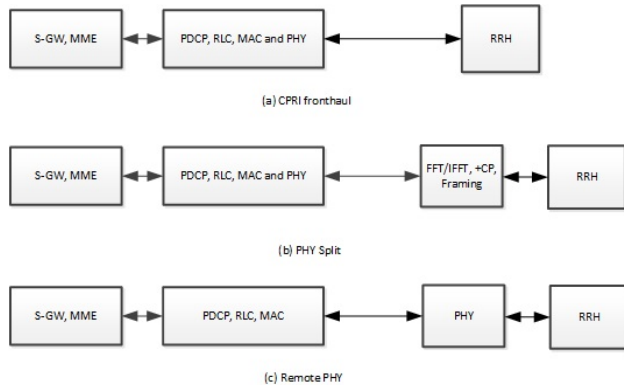


Fig. 3. Three Centralized Architecture Options

Despite all the benefits C-RAN promises, its key challenge lies in its fronthaul requirements which can become prohibitive in numerous practical deployment scenarios, e.g., when fiber as specified in [3] is not available to an operator deployment, especially with large number of antennas and high system bandwidth. To tackle this challenge, the original C-RAN concept evolved into different variations over the past three years.

Common Public Radio Interface (CPRI) is the standard interface connecting an RRH and the BBU pool, as shown in

Fig. 3(a). CPRI transmits the digitized I/Q samples per transmit antenna. The overall CPRI rate increases linearly with the number of transmit antennas and bandwidth. E.g., for a 20 MHz LTE system with 2 transmit antennas, the required CPRI rate is 2.46 Gbps. The CPRI rate can quickly go over 10 Gbps when 2- or 3-level carrier aggregation is deployed.

Various compression technologies are available targeting to compress the CPRI rate [3]. Those solutions typically achieve around 50% compression ratio. Further compression is a research topic, for instance the work presented in [4]. Motivated by the benefits of CPRI compression, RAN network re-architecture options are sought to balance the CPRI rate and the baseband signal processing at RRH.

One of the options closely connected with CPRI compression is that so-called PHY split option shown in Fig. 3(b). The baseband functionality is divided between the RRH and BBU pool. Some standard functions, such as FFT/IFFT, CP addition, and framing, are moved to the RRH side, with all other, typically more complex signaling processing functions such as channel estimation, MIMO detection and turbo decoding, are performed at the BBU pool. This simple re-architecture option reduces the fronthaul rate requirements by 50%. If on top of the described functionality at the RRH, we can also perform frequency-domain compression technologies, the fronthaul rate requirements can be further reduced to about 600 Mbps (for 20 MHz LTE, 2 antennas, medium network loading), thus enabling both wireless fronthaul and Ethernet fronthaul C-RAN deployments.

In addition to CPRI fronthaul and PHY split options, the remote PHY re-architecture option is shown in Fig. 3(c). Since all physical-layer processing functions are moved to the RRH, fronthaul rate requirement is reduced to the required maximum MAC throughput in this option, which is around 10% of the CPRI rate. The price paid for this reduction in the fronthaul requirements is the increased scheduling delay and longer HARQ turnaround time, which typically leads to reduced network performance as shown in Section III.

B. L2 locally coordinated: Anchor-Booster architecture

Anchor-Booster refers to an architecture envisioned for 5G networks which provided enhanced network performance when non-ideal backhaul/fronthaul links are used to connect macro eNBs with small cells or between small cells, see Fig. 4. The anchor-booster architecture is introduced to address three key challenges in HetNet deployments with non-ideal backhaul: Mobility robustness, per user throughput and signaling load due to frequent handovers. This architecture is an active study item in 3GPP Release 12.

TABLE I: RAN ARCHITECTURE OPTIONS COMPARISON

RAN Architecture Options		Centralized C-RAN			Anchor-booster	Edge integrated	
		CPRI	PHY Split	Remote PHY			
Fronthaul Rate		~2.5Gbps, ~1.25Gbps w/compression	~1Gbps, ~350Mbps w/ compression	~200Mbps	~200Mbps	~200Mbps	
Tolerable latency		~1-4 ms, Higher than latency reduce performance			Non-ideal		
Capacity Enhancement Techniques	Co-channel	ICIC/eICIC/FeICIC					
		DL/UL CoMP			DL CSCB, UL CoMP		Ongoing Rel-12 study items
	Non co-channel	Rel-11 Carrier Aggregation					

The anchor cell is mainly responsible for mobility, because the coverage of the anchor cell is larger than the one of the booster cell. As such, the UE does not need to handover to another cell as long as the UE is within the coverage of the anchor eNB. The booster eNB is mainly responsible for offloading data transmissions to and from the UE. Both the anchor and booster cells can serve packet data depending on the required QoS. For example, the anchor cell can serve delay sensitive data, e.g. VoIP, while the booster cell can serve non-delay sensitive data, e.g. FTP.

The anchor-booster architecture is not limited to LTE-based networks only. The booster cells are expected to operate based on different radio access technologies (RATs), e.g., WLAN or a new RAT, or based on different frequency bands, e.g., millimeter wave band, in order to provide high data rate and large offloading gains with reduced cost.

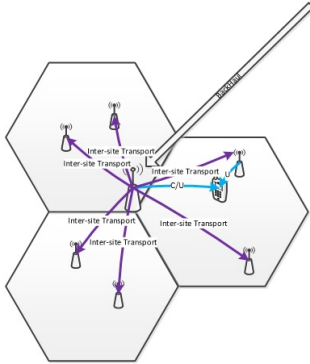


Fig. 4. L2 locally coordinated: Anchor-booster architecture

C. Fully integrated at the edge

The fully integrated at the edge architecture enables RAN level edge caching. Edge caching uses storage capability at the network edge (S-GWs, macro eNBs and small cell eNBs) for storing popular contents for fast retrieval and access. This architecture will alleviate the increased backhaul burden, reduce communication latency, improve user experience, and enable content-aware cross-layer optimization.

The fully integrated at the edge architecture is currently under discussion in NGMN (next generation mobile

networks). To realize this architecture, standardization would be required to ensure the deployment of interoperable solutions. Potential standardization work includes:

- Local bearer establishment in support of cached data delivery
- Mobility management for seamless handover and service continuity in using edge caching services
- Protocol stack support of transparent caching, or hierarchical caching at the core network and RAN

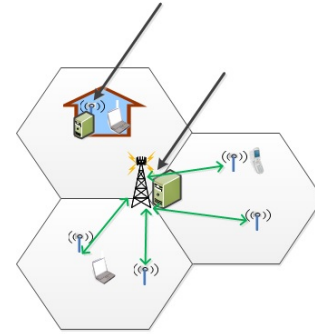


Fig. 5. Fully integrated at the edge

D. Summary of RAN re-architecture options

The different RAN architecture options discussed in this section are summarized in Table I with respect to their fronthaul rate requirement, tolerable latency, and typical applicable capacity enhancement techniques. Each architecture option has its advantages and limitations and typically fits into certain deployment scenario(s). Network function visualization provides an important tool to make network deployment equipment scalable and re-configurable to the different architectures and deployment scenarios. Before discussing network virtualization in more detail, Section III presents simulation results for the RAN architecture options of Section II as well as our vision for 5G network performance based on extensive evaluation studies.

III. PERFORMANCE RESULTS

This section we compare the performance results of the RAN architecture options described in Section II, i.e., ideal CRAN (referred to as CPRI fronthaul), PHY split, remote PHY and anchor-booster (non-ideal backhaul) using an LTE-based system. The reference system is an LTE Rel-10 based system

which does not use CoMP techniques (referred to as non-CoMP). Although the evaluation results in Table II refer to the LTE uplink operation, similar conclusions are drawn when the LTE downlink performance is considered. The CoMP Evaluation Methodology developed in 3GPP RAN1 for HetNet deployments is used according to LTE Rel-11 technical report TR 36.819 with the 1x2 antenna configuration, four pico cells per macro cell area and clustered user dropping [5]. The spectral efficiency (SE) results for both average and cell-edge user performance indicate that the PHY split architecture performs very close to the CPRI fronthaul, while significantly reducing the fronthaul requirements, see Table I. Therefore, the PHY split architecture would be the recommended one for 5G RAN architecture in the case the corresponding fronthaul rate requirements can be fulfilled; otherwise, the anchor-booster architecture seems to offer some performance benefits in addition to the rest of its advantages described in Section II.

Table II: System-level performance comparison of different RAN architecture options for the LTE uplink

	Average SE (b/s/Hz/cell)	Cell-edge user SE (b/s/Hz/user)
Non-CoMP	6.5 (0%)	0.088 (0%)
CPRI fronthaul	8.1 (+23%)	0.116 (32%)
PHY split	7.8 (+20%)	0.115 (31%)
Remote PHY	7.0 (+7%)	0.091 (3%)
Anchor- booster	6.8 (+3%)	0.089 (1%)

In order to provide insights into the benefits of C-RAN for 5G networks, Table III presents simulation results for both downlink (based on Joint Transmission CoMP) and uplink (based on Joint Reception CoMP) assuming the CPRI fronthaul architecture for different sizes of the coordination area which ranges from 1 macro cell (5 nodes) to 27 macro cells (135 nodes). The rest of the simulation assumptions are the same as the ones of Table II. According to the results in Table III, we can observe a remarkable increase of the C-RAN system spectral efficiencies, especially for the cell-edge user performance, with increasing size of the cooperation area. For 27-cell cooperation, said increase is on the order of 70% for the downlink and 122% in the uplink.

Setting out from the C-RAN performance with large size of the cooperation area (27-cell cooperation based on the results in Table III), Table IV presents evaluation results which can be considered representative for 5G networks in the sense that high level of cooperation for C-RAN is enabled, while the number of pico cells (small cells) per macro cell area increases to ten (compared to four for the reference system) and higher antenna configurations (8 antennas at the macro eNBs and 4 antennas at the pico eNBs) are deployed. Table IV contains the corresponding results for LTE Rel-10 (non-CoMP) and LTE Rel-11 (CoMP) according to the assumptions used for Table III. The results in Table IV should not be considered as apples-

to-apples comparison of technologies, but rather as comparison of representative 4G deployments based on LTE-Advanced and potential 5G deployments based on LTE RAN architecture evolution for systems beyond 2020. It is noted that the performance of 5G can be further enhanced with more advanced signal processing techniques, e.g., like the ones considered in massive MIMO systems [6], advanced anchor-booster architectures based on multi-RATs [7], new multiple access technologies [8], etc.

Table III: C-RAN system-level performance with different sizes of the coordination area

	Downlink		Uplink	
	Average SE (b/s/Hz/cell)	Cell-edge user SE (b/s/Hz/user)	Average SE (b/s/Hz/cell)	Cell-edge user SE (b/s/Hz/user)
Non-CoMP	9.5 (0%)	0.063 (0%)	6.3 (0%)	0.088 (0%)
CoMP, 1 cell	9.9	0.069	8.1	0.116
CoMP, 3 cells	10.0	0.078	8.2	0.122
CoMP, 9 cells	9.9	0.088	8.3	0.127
CoMP, 21 cells	9.7	0.100	8.7	0.164
CoMP, 27 cells	10.0 (5%)	0.108 (71%)	8.8 (40%)	0.195 (122%)

Table IV: System-level performance of 4G vs. 5G deployment scenarios

	Downlink		Uplink	
	Average SE (b/s/Hz/cell)	Cell-edge user SE (b/s/Hz/user)	Average SE (b/s/Hz/cell)	Cell-edge user SE (b/s/Hz/user)
LTE-A, Rel-10	9.5 (0%)	0.063 (0%)	6.3 (0%)	0.088 (0%)
LTE-A, Rel-11	10.0 (5%)	0.108 (71%)	8.8 (40%)	0.195 (122%)
LTE- based 5G	23.4 (146%)	0.314 (398%)	22.7 (260%)	0.581 (560%)

IV. NETWORK VIRTUALIZATION

As we discussed in the previous sections, network architecture is expected to play an increasingly pivot role in the future generations of wireless technologies. However, new network architecture can deliver the required cost and performance benefits only when it is implemented on high performance platforms. There are several limitations with the existing network implementations, including the long network innovation cycle, the high complexity and therefore high cost in introducing new network capabilities and services, the direct software dependency on the hardware, the rigid provisioning of compute resources that cannot adapt to the evolving traffic types, and the high complexity and cost in network management due to each network and often each network element's proprietary operations support system (OSS). Network virtualization is therefore increasingly used to address these limitations.

Network virtualization is currently implemented through two separate but related technology concepts: Software Defined Networking (SDN) and Network Functions Virtualization (NFV). The SDN technology was mainly developed by the IT industry, for efficient operation and management of datacenter and large IP and Ethernet networks. The goal of SDN is to separate the control plane from the data plane, and to make the control plan programmable through APIs to bring flexibility in how networks are deployed, operated and managed. The network function virtualization is primarily driven by network service providers. The goal of NFV is to virtualize network functions into software applications that can be run on industry off-the-shelf standard servers or as virtual machines running on those servers. Fig. 6 and Fig. 7 demonstrate the general architecture frameworks of SDN and NFV.

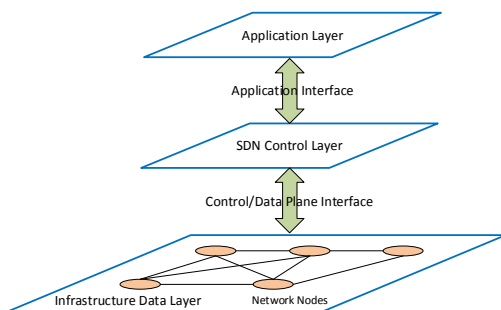


Fig. 6. SDN framework

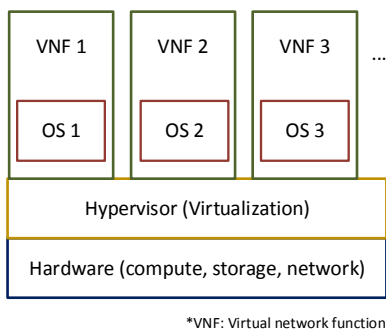


Fig. 7. NFV framework

Although SDN and NFV are two distinct initiatives, they share a lot of common characteristics and technologies. For example, an SDN controller implemented as a virtual function can be directly used in an NFV network. Intel has been a pioneer and industry leader in SDN and NFV technologies **Error! Reference source not found.** To be applied in carrier-grade networks, both SDN and NFV protocol design need to ensure scalability, reliability, quality of service and service management. For NFV, as conventional single-purpose network nodes are being virtualized and running on a single general-purpose computation platform, technical challenges include choosing the virtualization granularity, abstracting and logically partitioning of the physical resource at the virtualization layer (hypervisor), and meeting the real-time

processing requirements by the general-purpose computation platform.

Network virtualization benefits future 5G networks in many aspects. As we have demonstrated in previous sections, future network platforms need to be highly flexible and economical to accommodate different network architecture options. More importantly, network virtualization enables future 5G networks to efficiently scale in terms of data rate, number of users, applications and the level of network cooperation. Finally, network virtualization is essential for telecom and information technology convergence and integration. It is a foundation for future intelligent wireless networks that provide context, storage and processing for 5G mobile devices.

V. CONCLUSION

In this paper, we discuss the main RAN architecture options considered for 5G network evolution. Different deployment scenarios will favor different architecture option for cost and performance tradeoff. Representative simulation results are presented to demonstrate the differences among the different architecture options. We also provide insights into our vision of 5G network performance based on extensive simulation results. Finally, we argue that network virtualization, through the cooperation of software defined networking and network function virtualization, is required both from a technology and cost points of view in order to enable implementations of the RAN architecture options for 5G networks.

REFERENCES

- [1] Cisco visual networking index: Global mobile data traffic forecast update, 2012.
- [2] China Mobile Research Institute, "C-RAN: The road toward green RAN", White paper, 2011.
- [3] Common Public Radio Interface (CPRI), Interface specification, V6.0, August 2013.
- [4] CPRI compression transport for LTE and LTE-A signal, IEEE CHINACOM 2012, pp. 843-849, 2012.
- [5] 3GPP TR 36.819, Coordinated multi-point operation for LTE physical aspects, 2013.
- [6] H. Shirani-mehr, M. Sajadieh, and Q. Li, "Practical downlink transmission scheme for future LTE systems with many base station antennas," to appear in Proc. IEEE GLOBECOM 2013, Atlanta, USA, 2013.
- [7] Q. Li, H. Niu, G. Wu and R. Q. Hu, "Anchor-booster based heterogeneous networks with mmWave capable booster cell," to appear in Proc. IEEE GLOBECOM 2013, Atlanta, USA, Dec. 2013.
- [8] Y. Saito, A. Benjebbour, Y. Kishiyama, and T. Nakamura, "System-level performance evaluation of downlink non-orthogonal multiple access (NOMA)," in Proc. IEEE PIMRC 2013, pp.1-5, London, 2013.
- [9] John Healy and Cliff Farrah, "On the promise of an SDN: A white paper", Intel and Beacon Group White Paper, Spring 2013, url: <http://www.adaranet.com/pdf/PR-20130603.pdf>